

A 2026 Literature-Implied Partial Answer to the Compact-Open Fuglede Problem in $\mathbb{Q}_p^2$

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Status and exact source question

This is a `literature_implied_answer` (partial subcase) packet. Fan–Fan–Liao–Shi end Section 6.4 of [1], on printed and PDF page 23, with the statement

“Fuglede’s conjecture in $\mathbb{Q}_p^2$ is open, even under the compact open assumption.”

The one-dimensional assertion in the paper’s title is proved in that same paper and is not the target here.

Zhou’s later Theorem 1 [2], on PDF page 26, proves that tiling and spectral sets coincide in $\mathbb{Z}_{p^2} \times \mathbb{Z}_{p^2}$. The supporting paper does not identify this as a partial answer to the quoted p -adic question. The identification is made below through an exact finite-quotient reduction.

Finite-quotient identification

Normalize Haar measure by $|\mathbb{Z}_p^2| = 1$, and use the standard additive character χ of $\mathbb{Q}_p$ whose kernel is $\mathbb{Z}_p$. Put

$$K = \mathbb{Z}_p^2, \quad H = p^m K, \quad G_m = K/H \simeq (\mathbb{Z}/p^m \mathbb{Z})^2.$$

For $A \subset G_m$, choose lifts in K and write

$$\Omega_A = A + H.$$

Proposition 1 (exact compact-open reduction). *The set Ω_A tiles $\mathbb{Q}_p^2$ by translations if and only if A tiles G_m . It is spectral in $\mathbb{Q}_p^2$ if and only if A is spectral in G_m for the pairing*

$$(a, s) \mapsto \chi(p^{-m} \langle a, s \rangle).$$

Proof. Let L be a set of representatives for $\mathbb{Q}_p^2/K$. If $A \oplus B = G_m$, then

$$\Omega_A + (B + L) = \mathbb{Q}_p^2$$

is a disjoint tiling: first choose the unique K -coset, then the unique decomposition modulo H . Conversely, suppose $\Omega_A + T$ tiles $\mathbb{Q}_p^2$. Every translate $\Omega_A + t$ is contained in $t + K$. Restrict the tiling to one coset $\ell + K$ and reduce the relevant translations $t - \ell$ modulo H . Two of them cannot have the same residue, because the corresponding translates would coincide and overlap on a set of positive measure. The resulting residue set B therefore satisfies $A \oplus B = G_m$.

For spectrality, note that

$$H^\perp = \{\xi : \chi(\langle \xi, h \rangle) = 1 \text{ for all } h \in H\} = p^{-m}K.$$

Let Γ represent $\mathbb{Q}_p^2/H^\perp$. If S is a finite spectrum of A , then

$$\Lambda = \bigcup_{\gamma \in \Gamma} (\gamma + p^{-m}S) \tag{1}$$

is a spectrum of Ω_A . Indeed, frequencies belonging to different $H^\perp$ -cosets are orthogonal on every H -ball. Within a fixed coset, their inner products are

$$|H| \sum_{a \in A} \chi(p^{-m}\langle a, s - s' \rangle),$$

which are precisely the finite Fourier inner products on A . Completeness follows by the Fourier decomposition on the compact group H : each γ -sector is a copy of the $|A|$ -dimensional space $\ell^2(A)$.

Conversely, partition any spectrum Λ of Ω_A into its cosets modulo $H^\perp$. The same orthogonal sector decomposition shows that each coset must contain exactly $|A|$ frequencies. Fixing one frequency in a coset, the remaining differences lie in $H^\perp/K \simeq G_m$; after removing one common diagonal phase from the cell vectors, those $|A|$ residues form an orthogonal basis of characters on A . Thus A is spectral in G_m . $\square$

Every compact-open subset of $\mathbb{Q}_p^2$ can, after a translation and a p -adic dilation, be put in the form Ω_A . Its *depth* is the least such m . Translation and dilation preserve both sides of Fuglede's equivalence.

Partial answer

Theorem 1. *Let $\Omega \subset \mathbb{Q}_p^2$ be compact and open.*

1. *If Ω has depth at most two, then Ω is spectral if and only if it tiles $\mathbb{Q}_p^2$.*
2. *At arbitrary depth m , after normalization $\Omega = \Omega_A \subset K$, tiling implies spectrality when $|A| = p$ or $|A| = p^{2m-1}$. At the latter cardinality the converse also holds.*
3. *At arbitrary depth, the equivalence holds whenever A has a subgroup tiling complement. Equivalently for the reverse formulation, it holds when A has a subgroup spectrum. This includes all residue sets that are graphs over either coordinate.*

The unrestricted compact-open problem, equivalently the assertion for every subset of G_m and every $m \geq 3$, remains open.

Proof. For (1), represent a depth-at-most-two set at level two and combine Proposition 1 with Zhou's Theorem 1. For (2), use Proposition 1 and Zhou's Lemmas 11, 13, and 14 [2, PDF pp. 11–12]. Lemma 14 says in particular that a spectral set with $p^{2m} > |A| \geq p^{2m-1}$ must have size p^{2m-1} and tile.

For (3), if $A \oplus H = G_m$ with $H \leq G_m$, then A is a transversal for H and the annihilator $H^\perp$, of cardinality $|A|$, is a spectrum of A : its character table is the Fourier table of G_m/H . Conversely, if a subgroup S is a spectrum of A , orthogonality and $|S| = |A|$ force A to be a transversal for $S^\perp$, so $A \oplus S^\perp = G_m$. A graph over one coordinate is a transversal for the subgroup in the other coordinate. $\square$

Why the evident induction does not finish the problem

Projection $G_m \rightarrow G_{m-1}$ does not preserve tiling as a property of the set-valued image. For example, in $(\mathbb{Z}/8\mathbb{Z})^2$ take

$$A = \{(x, f(x)) : x \in \mathbb{Z}/8\mathbb{Z}\}, \quad f(4) = 1, \quad f(x) = 0 \ (x \neq 4).$$

This is a tile with vertical subgroup complement, but its image modulo four has the five points

$$(0, 0), (0, 1), (1, 0), (2, 0), (3, 0),$$

and cannot tile the 16-element group $(\mathbb{Z}/4\mathbb{Z})^2$. An induction must retain nonconstant fiber multiplicities and their Fourier phases; the level- $m - 1$ set alone loses them. The general- m lemmas in [2] control only extremal sizes and do not bridge these intermediate fiber patterns.

Provenance, search bounds, and review recommendation

The run's cheap indexes were searched for arXiv:1512.08904, its title, the quoted sentence, and the core finite-group terms. A bounded current arXiv/web search used “ $\mathbb{Q}_p^2$ Fuglede,” “ $\mathbb{Z}_{p^m} \times \mathbb{Z}_{p^m}$ spectral tile,” rank-two p -groups, arXiv:2412.18132, and its companion arXiv:2412.16865. No full all- m theorem or counterexample was found. The strongest direct result was the 22 March 2026 revision of [2], whose introduction itself describes rank-two results as limited and states only the $m = 2$ theorem plus the cited general- m lemmas.

Three upgrade routes were audited: quotient induction (obstructed by the explicit graph example), subgroup-complement reduction (complete only for that structural class), and cardinality bootstrapping (leaving intermediate sizes). Thus this packet does not claim a full solution. Human review is recommended, with primary attention to the converse spectral reduction in Proposition 1; confidence in the stated partial result is high.

References

- [1] A. Fan, S. Fan, L. Liao, and R. Shi, *Fuglede's conjecture holds in $\mathbb{Q}_p$* , arXiv:1512.08904v1 (2015), Section 6.4, p. 23, arXiv:1512.08904.
- [2] W. Zhou, *On Tiling and Spectral Sets in $\mathbb{Z}_{p^2} \times \mathbb{Z}_{p^2}$* , arXiv:2412.18132v4 (22 March 2026), Theorem 1 and Lemmas 11, 13, 14, arXiv:2412.18132.